

Program : Diploma in Renewable Energy	
Course Code : 5297	Course Title: Electrical Machines Lab II
Semester : 5	Credits: 1.5
Course Category: Program core course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To comprehend the performance of synchronous machines.
- To understand synchronization of alternators.
- To comprehend the performance of induction machines.
- To understand the starting methods of induction motors.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Machines		AC Machines Lab 1	4

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop the performance characteristics of three phase induction motors	12	Applying
CO2	Develop the no load performance of a synchronous generator	12	Applying
CO3	Construct the performance characteristics of synchronous machines	9	Applying
CO4	Identify the performance of single phase Induction motors	6	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Develop the performance characteristics of three phase induction motors		
M1.01	1. Identify and collect the name plate details of induction motor 2. Identify the parts of a squirrel cage and slip ring induction motors. 3. Familiarize starters for three phase induction motors.	3	Applying
M1.02	Start a three phase induction motor and determine its slip at various loads.	3	Applying
M1.03	Conduct load test on three phase induction motor and plot various curves .	3	Applying
M1.04	Draw the equivalent circuit of a three phase induction motor by conducting no load and blocked rotor tests	3	Applying
CO2	Develop the no load performance of a synchronous generator		

M2.01	1. Identify and collect the name plate details of synchronous generator. 2. Identify the parts of synchronous generator.	3	Applying
M2.02	Run an alternator at different speeds with constant field current and measure the frequency and voltage in each case.	3	Applying
M2.03	Pre-determine regulation of three phase alternator by EMF method.	3	Applying
M2.04	Pre-determine regulation of three phase alternator by MMF method.	3	Applying
	Lab Test – I	3	
CO3	Construct the performance characteristics of synchronous machines		
M3.01	Determine the regulation of an alternator at various power factors by conducting direct load test .	3	Applying
M3.02	Synchronize a three phase alternator with existing power supply using various methods (dark lamp / bright lamp / synchroscope)	3	Applying
M3.03	Construct the V and Inverted V curves of synchronous motor	3	Applying
CO4	Identify the performance of single phase Induction motors		
M4.01	1. Identify the parts and terminals of a single phase induction motor 2. Run a single phase induction motor in forward and reverse directions.	3	Applying
M4.02	Develop the performance characteristics of a single phase induction motor by conducting load test	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	BL Theraja. Electrical technology. Vol- II: S Chand & co.
T2	J.B. Gupta. Theory & Performance of Electrical Machines: S. K. Kataria & Sons
R3	Electrical Machinery : Dr. P.S. Bimbhra

Online Resources

Sl.No	Website Link
1	http://em-coep.vlabs.ac.in/
2	Virtual Lab http://vlabs.iitb.ac.in/vlab/labsee.html
3	http://em-coep.vlabs.ac.in/List%20of%20experiments.html?domain=Electrical%20Engineering
4	http://vem-iitg.vlabs.ac.in/
5	https://www.ee.iitb.ac.in/course/~emlab/lab-manual.html

Student Activity**Suggested Open-ended Experiments:**

Students can do open ended experiments as a group of 3-5 preferably a simulation

work. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open ended experiment of their choice.